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USER MANUAL

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Automated Synchrotron Preprocessing Pipeline (ThePrae)

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This tool is a high-performance, deterministic data reduction and analytical execution environment engineered specifically to eliminate operator-dependent free parameters and manual subjectivity in large-scale X-ray Absorption Spectroscopy (XAS) workflows. Developed directly to meet the high-throughput demands of modern automated synchrotron beamlines—such as the multi-file acquisition systems at the Synchrotron Light Research Institute (SLRI), Thailand—the software bridges the critical gap between raw experimental data logs and rigorous, reproducible spectroscopic analysis. By decoupling baseline normalization and energy calibration outcomes from user-defined manual inputs, this framework guarantees absolute execution consistency: identical spectral data arrays will always generate identical numerical outputs from raw spectroscopic files alone.

The Core Preprocessing & Calibration Framework operates through an orchestrated suite of six discrete, specialized computational blocks designed to mitigate mechanical distortions and monochromator thermal drift profiles via systematic, rule-based conditioning algorithms:

1. **Time-Mapped External Calibration Module:** Synchronizes external timestamped calibration reference points with primary experimental data arrays to baseline global temporal energy grids.
2. **Absolute Energy Calibration or Direct Energy Calibration:** Executes real-time, parallel deconvolution of reference foil spectra acquired concurrently with target sample collection, stabilizing spatial data shifts.
3. **Nearest-Foil Block Calibration Module:** Applies adaptive spatial-temporal interpolation algorithms to map the closest uncorrupted calibration references across discrete experimental batches.
4. **Scan-to-Scan Energy Alignment or Relative Energy Alignment via in-line Reference:** Performs a multi-pass hierarchical adjustment to anchor non-linear derivative energy shifts, safeguarding global energy alignment against anomalies.

5. **Spectral Alignment Module (Pre-Merge Synchronization):** Enforces rigid cross-spectrum normalization and grid synchronization on individual spectral arrays prior to final accumulation, preventing feature distortion.
6. **Multi-Scan Ensemble Merging Module:** Implements advanced deterministic blending and artifact-rejection filtering over multi-scan series to maximize data signal-to-noise ratio (SNR).

Through this decentralized, multi-modular architecture, the pipeline automates high-precision energy anchoring and real-time synchronization tracking, resulting in synchronized datasets optimized for downstream analytical workloads.

The Persistent Reconstruction of Absorption Ensembles

SLRI-Cal Preview Spectra

Initialize Calibration Plot selected file Export Normalized Data Derivative selected file Export Derivative Data

Calibration Mode : External Calibration

Element : Auto Select

E0 [eV]: Auto Select

Calibration Option

Energy column : Auto Select

$\mu(E)$ column : Auto Select

Nearest Foil Mapping

Select Reference Match to Nearest Foil

Align & Merge

Select Reference Align Sample Merge Sample

Reference $\mu(E)$ column : Auto Select

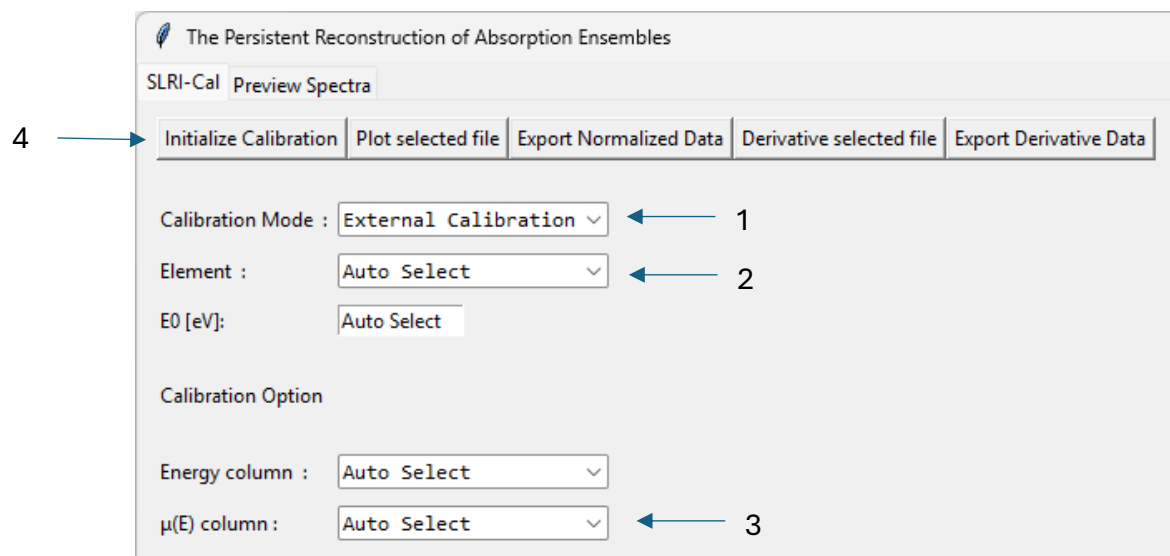
Sample $\mu(E)$ column : Auto Select

Input directory :

Output directory :

1. Time-Mapped External Calibration Module

The Time-Mapped External Calibration framework is explicitly engineered to dynamically decouple continuous synchrotron monochromator thermal energy drifts using physical experimental Unix timestamps logged directly at reference nodes.



Step-by-Step Execution Guide:

- **Step 1:** Navigate to the Calibration Mode settings framework and select "**External Calibration**" from the operational configuration menu list.
- **Step 2:** Choose your measured transition element from the built-in "**Element**" selection dropdown menu. If your specific absorption edge position $E0$ is absent from the standard library reference list, you can input the explicit theoretical $E0$ value (in eV) directly into the provided custom input text parameter grid.
- **Step 3:** Define the specific column data path by selecting the correct "**mu(E) column of external reference**" parameter from the channel selector.
- **Step 4:** Click the "**Initialize calibration**" operational execution button. A system directory browser window will emerge; select the baseline destination folder containing your raw un-normalized spectrum trace files.

Strict Target Directory Requirements:

To execute this automated time-mapping data cascade, the target execution folder must contain a correlated blend of actual sample scans alongside baseline reference foil files. The processing engine scans filenames adaptively; reference standards must explicitly contain at least one of the following identifying structural string tokens within their filenames (case-insensitive): **"foil", "ref", "std", "metal", "standard", or "calibration"**.

Mandatory File Header Metadata Specifications:

The continuous Unix timestamp mapping engine requires explicit date and duration markers embedded within the top header blocks of every raw data file. Currently, the operational parser natively decodes the following structural variations:

- *Accepted Date Signatures:*
 - #Experiment date: 02 September 2025
 - #Experiment date: Tuesday, July 8, 2025
- *Accepted Time Signatures:*
 - #Duration: 09:43:00 - 09:46:21
 - #Duration: 1:23:24 AM - 1:42:37 AM

Developmental Notice: Alternative string parsing formats for experimental date/time signatures are currently under active development. If your specific synchrotron beamline utilizes a discrete header standard not captured above, please transmit sample source data files directly to the core development team at **anan@slri.or.th** for custom routine onboarding integration.

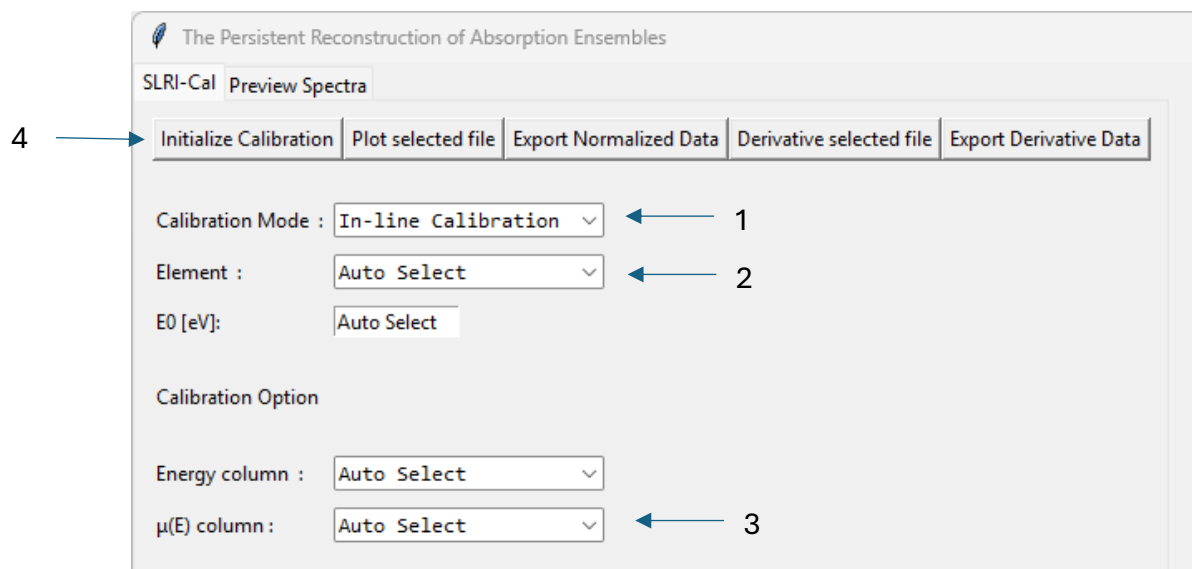
Data Output Infrastructure & Naming Conventions:

Upon successful termination of the optimization pipeline, *it* automatically secures and isolates the energy-rectified output data structures:

1. All calibrated files are automatically exported into a newly spawned sub-folder designated as **"External_Cal"** within your root target directory.
2. The runtime engine systematically preserves file identification traceability by appending the string suffix **"_cal"** directly to the terminal end of each modified output file name (e.g., sample.txt becomes sample_cal.txt).

2. Absolute Energy Calibration or Direct Energy Calibration

The Simultaneous In-line Calibration framework operates by automatically localizing the mathematical inflection point (the maximum intensity of the first derivative) and anchoring it precisely onto the designated theoretical target energy (E_0) threshold grid.



Step-by-Step Execution Guide:

- **Step 1:** Navigate to the Calibration Mode configuration panel and select "**In-line Calibration**" from the operational dropdown listing framework.
- **Step 2:** Choose your measured transition element from the built-in "**Element**" library selection menu. If your specific absorption edge position (E_0) is absent from the standard reference table, you can explicitly input the custom theoretical E_0 value (in eV) directly into the dedicated text input parameter grid.
- **Step 3:** Define the specific column data path by selecting the correct "**mu(E) column of in-line reference**" parameter from the channel selector.
- **Step 4:** Click the "**Initialize calibration**" operational execution button. A directory browser window will emerge; select the target baseline folder containing your raw un-normalized spectroscopic transmission files.

Strict Dataset Directory Requirements:

To guarantee error-free automation execution loops across the entire dataset, **every individual raw data file** situated within the targeted execution folder must rigidly contain the continuous in-line reference channel embedded inside the exact same column index coordinate.

Data Output Infrastructure & Naming Conventions:

Upon successful termination of the optimization execution cascade, *it* automatically exports and isolates the energy-rectified output data structures:

1. All calibrated output spectrum files are automatically generated and exported into a newly spawned sub-folder designated as "**Inline_Cal**" within your root target directory folder.
2. The software core systematically preserves full data traceability by appending the string suffix "**_cal**" directly to the terminal end of each modified output file name (e.g., data_001.txt becomes data_001_cal.txt).

Operational Engineering Tip (Reference-Only Ingestion Routing):

In specific experimental contexts where you need to perform automated energy calibration exclusively on folders that contain **pure reference scans alone** (without interleaved unknown samples), you can seamlessly exploit this in-line module. To execute this routing setup, simply target and select the primary absorption channel column index ($\mu(E)$ column) of your reference standard as the processing focus parameter, and the runtime engine will handle the global alignment automatically.

3. Nearest-Foil Block Calibration Module

The Nearest-Foil Block Calibration framework is engineered to execute traditional block-wise geometric spectrum adjustments, enabling automated mapping and alignment of massive sample ensembles directly toward targeted standalone reference foil standard files.

The screenshot displays the 'Nearest-Foil Block Calibration Module' interface. It features two main sections: 'Nearest Foil Mapping' and 'Align & Merge'. The 'Nearest Foil Mapping' section contains two buttons: 'Select Reference' (pointed to by callout 2) and 'Match to Nearest Foil' (pointed to by callout 4). The 'Align & Merge' section contains three buttons: 'Select Reference', 'Align Sample', and 'Merge Sample'. Below these buttons are two dropdown menus. The first dropdown, labeled 'Reference $\mu(E)$ column:', is set to 'Auto Select' and is pointed to by callout 1. The second dropdown, labeled 'Sample $\mu(E)$ column:', is also set to 'Auto Select' and is pointed to by callout 3.

Step-by-Step Execution Guide:

- **Step 1:** Define your reference matrix channel by selecting the correct absorption column parameter index from the "**reference $\mu(E)$ column**" channel selector dropdown menu.
- **Step 2:** Click the "**select reference**" operational execution button. A system browser window will emerge; select the specific, master reference standard foil data file that your samples will align toward.
- **Step 3:** Define your target sample channel configuration by choosing the corresponding data path from the "**sample $\mu(E)$ column**" selection dropdown menu (this channel is explicitly utilized for interactive graphical trace rendering updates).
- **Step 4:** Click the "**Match to Nearest foil**" operational execution button. Select the baseline directory or individual sample spectrum data files that require programmatic mapping and optimization toward your previously loaded reference node.

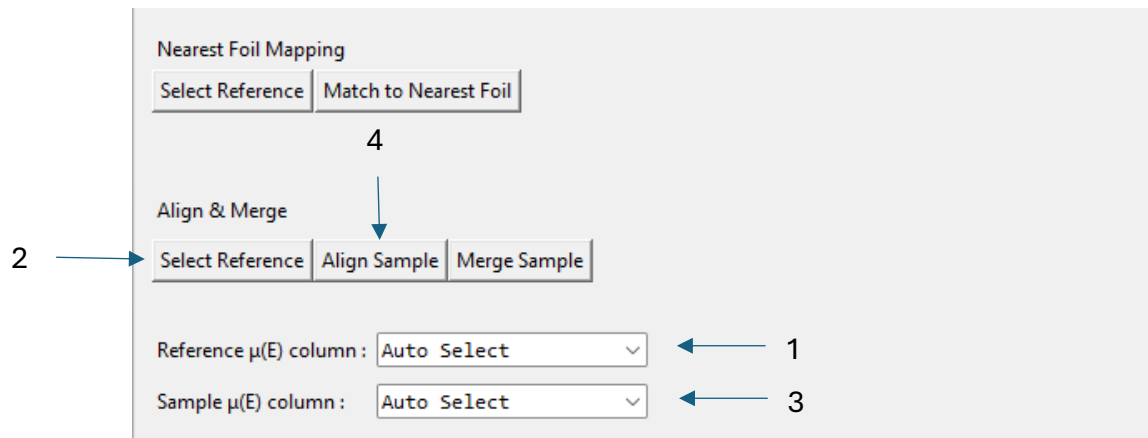
Data Output Infrastructure & Naming Conventions:

Upon successful termination of the array synchronization pipeline, *it* automatically secures, partitions, and exports the energy-rectified output data structures:

1. All calibrated output spectrum files are automatically generated and exported into a newly spawned sub-folder designated as "**Nearest_Foil_Mapping**" located within your root target directory.
2. The runtime engine systematically preserves full database and file traceability by appending the string suffix "**_nfm**" directly to the terminal end of each modified output filename (e.g., unknown_spec_001.txt becomes unknown_spec_001_nfm.txt).

4. Scan-to-Scan Energy Alignment or Relative Energy Alignment via in-line Reference

The Two-Stage Secondary Alignment framework is a high-performance signal-matching engine engineered specifically to execute advanced sample-to-reference vector trace adjustments, aligning the embedded in-line channel of unknown samples directly toward pre-calibrated reference masters.



Step-by-Step Execution Guide:

- **Step 1:** Define your master baseline channel by selecting the corresponding parameter from the "**reference $\mu(E)$ column**" selection dropdown menu list.
- **Step 2:** Click the "**select reference**" operational execution button. A system directory browser window will emerge; select the specific, energy-rectified reference data file that has already undergone prior energy calibration modifications.
- **Step 3:** Define your incoming alignment channel by navigating to the "**Sample $\mu(E)$ column**" dropdown menu, and explicitly choose the specific column index containing the **in-line reference channel data** of your raw sample spectra.
- **Step 4:** Click the "**Align Sample**" operational execution button. Select the baseline directory or batch sequence of sample files that require precise programmatic alignment and geometric tuning toward your pre-calibrated reference node.

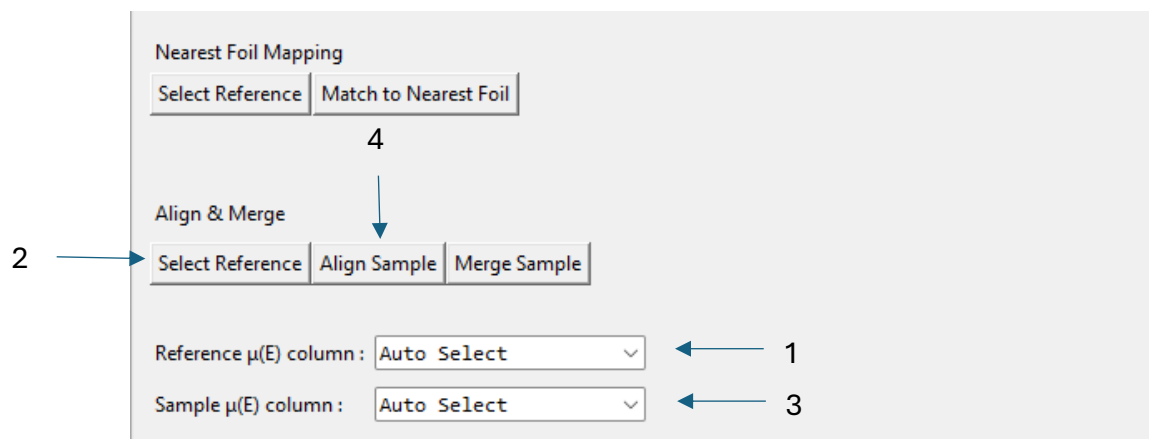
Data Output Infrastructure & Naming Conventions:

Upon successful termination of the dual-stage numeric synchronization pipeline, *it* automatically exports and secures the energy-rectified array outputs:

1. All aligned output spectrum data structures are automatically generated and exported into a newly spawned sub-folder designated as "**Aligned_Results**" within your root target directory folder.
2. The runtime engine systematically preserves full data traceability across the processing queue by appending the string suffix "**_aln**" directly to the terminal end of each modified output file name (e.g., sample_matrix_001.txt becomes sample_matrix_001_aln.txt).

5. Spectral Alignment Module (Pre-Merge Synchronization)

The Spectral Alignment framework operates identically to the structural architecture of the Two-Stage Secondary Alignment engine, but is specialized explicitly for target sample-to-sample trace synchronization to guarantee seamless error-free data integration prior to multi-scan ensemble merging operations.



Step-by-Step Execution Guide:

- **Step 1:** Select the core reference channel index from the "**reference $\mu(E)$ column**" selection dropdown menu list.
- **Step 2:** Click the "**select reference**" operational execution button and load the specific target reference scan data file. **Strict Physical Requirement:** This designated master reference scan *must* have already successfully undergone prior energy calibration modifications via the primary baseline calibration pipeline to preserve absolute tracking precision.
- **Step 3:** Navigate to the "**Sample $\mu(E)$ column**" selection dropdown menu and explicitly choose the **exact same column index** as defined in Step 1. Because this pipeline aligns home-group spectral traces toward their calibrated cohort master, the reference and target channel indices must tightly mirror each other inside the execution RAM arrays.
- **Step 4:** Click the "**Align Sample**" operational execution button. Select the batch folder or multiple spectrum data files of identical chemical systems that require precise array alignment and geometric stabilization before triggering final averaging combinations.

Data Output Infrastructure & Naming Conventions:

Upon successful termination of the localized sample array synchronization loop, *it* automatically exports and secures the energy-rectified output data structures:

1. All aligned spectrum files are systematically generated and exported into the designated spawned sub-folder named **"Aligned_Results"** within your root target directory folder.
2. The runtime engine preserves file database traceability by appending the string suffix **"_aln"** directly to the terminal end of each modified output filename (e.g., batch_scan_001.txt becomes batch_scan_001_aln.txt).

6. Multi-Scan Ensemble Merging Module

The Multi-Scan Ensemble Merging framework is a high-throughput averaging engine engineered to collapse multiple aligned spectrum scans of identical chemical systems into a single, high-signal-to-noise ratio combined data structure, preparing pristine unified inputs optimized for downstream linear combination optimizations.

The screenshot shows a software interface for the Multi-Scan Ensemble Merging Module. It is divided into two main sections: "Nearest Foil Mapping" and "Align & Merge".

Nearest Foil Mapping: This section contains two buttons: "Select Reference" and "Match to Nearest Foil".

Align & Merge: This section contains three buttons: "Select Reference", "Align Sample", and "Merge Sample". A blue arrow labeled "2" points to the "Merge Sample" button.

Below the buttons are two dropdown menus:

- "Reference $\mu(E)$ column:" with a dropdown menu showing "Auto Select".
- "Sample $\mu(E)$ column:" with a dropdown menu showing "Auto Select". A blue arrow labeled "1" points to this dropdown menu.

Step-by-Step Execution Guide:

- **Step 1:** Define the core target spectral data path by selecting the correct parameter index from the **"Sample $\mu(E)$ column"** selection dropdown menu.
- **Step 2:** Click the **"Merge Sample"** operational execution button. A system browser window will emerge; select the full batch sequence or multiple previously aligned spectrum files (containing the tracking suffix **_aln**) that require algorithmic average blending.

Data Output Infrastructure & Naming Conventions:

Upon successful termination of the array summation and mathematical averaging loops, it automatically exports and secures the unified output data structures:

1. The combined merge result files are systematically generated and exported into a newly spawned sub-folder designated as "**Merge_Results**" within your root target execution directory.
2. The runtime engine rigidly preserves historical processing traceability across your spectral database by appending the string suffix "**_merge**" directly to the terminal end of the combined output filename (e.g., zinc_crystal_batch.txt becomes zinc_crystal_batch_merge.txt).

SLRI-LCF User Manual

The Persistent Reconstruction of Absorption Ensembles

This guide describes how to operate the SLRI-LCF application to determine the composition of unknown samples using Linear Combination Fitting (LCF).

The screenshot displays the SLRI-LCF application window. The title bar reads "The Persistent Reconstruction of Absorption Ensembles". The main window has a tab labeled "SLRI-LCF". Below the tab is a menu bar with four buttons: "Lower (eV)", "Upper (eV)", "Import Target file", "Run LCF Analysis", "Export Fitting Result...", and "Reset".

The main area contains several input fields and buttons:

- Synthetic Spectra:** A text input field with the value "50000".
- Element:** A dropdown menu with "Auto Select" selected.
- Energy column:** A dropdown menu with "Auto Select" selected.
- Target $\mu(E)$ column:** A dropdown menu with "Auto Select" selected.
- Reference 1 $\mu(E)$ column:** A dropdown menu with "Auto Select" selected, followed by an "Import RF Group 1" button and a "Result" button.
- Reference 2 $\mu(E)$ column:** A dropdown menu with "Auto Select" selected, followed by an "Import RF Group 2" button and a "Result" button.
- Reference 3 $\mu(E)$ column:** A dropdown menu with "Auto Select" selected, followed by an "Import RF Group 3" button and a "Result" button.
- Reference 4 $\mu(E)$ column:** A dropdown menu with "Auto Select" selected, followed by an "Import RF Group 4" button and a "Result" button.
- Reference 5 $\mu(E)$ column:** A dropdown menu with "Auto Select" selected, followed by an "Import RF Group 5" button and a "Result" button.

Each reference section has a large empty text area below the buttons, likely for displaying the result of the fitting process.

1. Target Sample Import (Unknown)

- **Import Target File:** Click the [Import Target file] button on the top menu bar to load the spectrum file of your unknown sample.

- Element: Select the target element from the dropdown menu to automatically define the absorption edge energy E_0 for your sample.
- Energy column: Select the specific column in your target file that contains the energy data.
- Target $\mu(E)$ column: Select the column in your target file that contains the absorption coefficient $\mu(E)$ data.

2. Reference Data Import (Standards)

- Import RF Groups (1 to 5): You can load reference standards into up to 5 distinct groups. Each group supports bulk importing of more than one file.
- Reference $\mu(E)$ column: For each imported group, select the appropriate column that contains the reference $\mu(E)$ data from the dropdown menu.

3. LCF Analysis & Performance Settings

- Synthetic Spectra: Enter your desired number of simulated spectrum combinations in the field. The software will generate and match these variations against your target.
- High-Speed Processing: The application features a high-performance matching algorithm. It can evaluate up to 50,000 synthetic mixtures in just 5.23 seconds (benchmarked on a standard Intel Core i5-14500 personal computer).
- Run LCF Analysis: Click the [Run LCF Analysis] button on the top menu to start the fitting process. The algorithm evaluates the mixtures to find the best match with the lowest R-factor, identifying exactly which reference compounds are present and their respective proportions.

4. Utilities & Exporting Results

- Result: The calculated proportions and fitting details for each reference will be displayed in the text boxes next to and below each respective group.
- Lower (eV) / Upper (eV): Adjust these fields on the top menu bar to define the specific energy window (XANES region) for the LCF calculation.
- Export Fitting Result...: Click this button to save the statistical summaries, proportions, and fit data into an external report file.
- Reset: Click [Reset] to clear all current inputs, imported files, and parameters to start a new analysis.

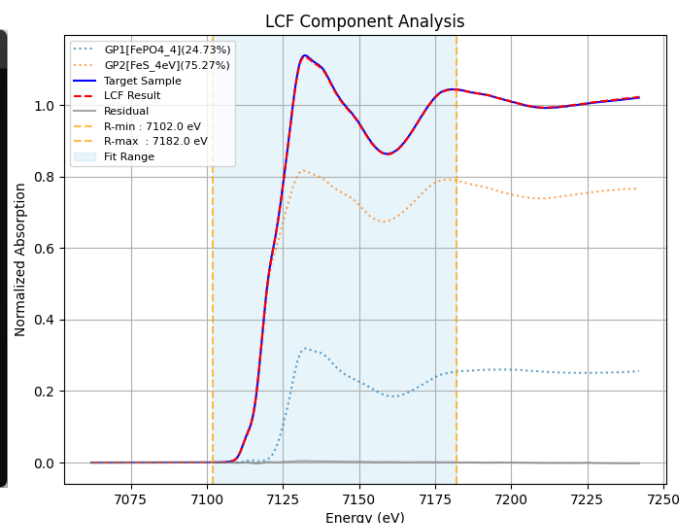
```
SLRI-Lcf
Target: FePO4_4eV_WB U8.5 L-4.5_EXAFS.1231_merge.txt
Target Sample loaded. Shape: (1, 17999)
Reference Standards GP1 loaded. Shape: (1, 17999)
Reference Standards GP2 loaded. Shape: (3, 17999)
Fitting Process: [ 100.00 % ] | 50,000 / 50,000 | R-factor : 0.00000462

Fit Range : [-10, 70] eV

Fitted Fractions:
GP1 (FePO4_4) : 0.2473 (24.73%)
GP2 (FeS_4eV) : 0.7527 (75.27%)

R-factor: 0.00000461
Chi-square: 0.002693
Reduced Chi-square: 0.000003
Degrees of Freedom: 800

Analysis Complete!
execution time: 4.592000 seconds
```



LCF Execution Results & Graphical Interpretation

This section demonstrates the program output and visual analysis using a generated ground-truth validation sample.

1. Data Loading & Validation Setup

- Target Sample: The analyzed target file is FePO4_4eV_WB U8.5 L-4.5_EXAFS.1231_merge.txt with a data shape of (1, 17,999) points. For software validation, this file functions as a ground truth test sample mathematically merged from 1 pure FePO₄ spectrum and 3 pure FeS spectra to enforce a pre-defined 1:3 ratio (25.00% : 75.00%).
- Reference Standards:
 - GP1 (Group 1): 1 reference standard loaded with a shape of (1, 17,099).
 - GP2 (Group 2): 3 reference standards loaded with a shape of (3, 17,099).

2. Fitting Constraints & Performance

- Fit Range: The LCF algorithm evaluated the data within an energy window of [-10, 70] eV relative to the absorption edge (E_0). On the plot, this specific calculation window is highlighted by the light-blue shaded region, bounded by orange dashed lines at R-min (7102.0 eV) and R-max (7182.0 eV).
- Fitting Process: Completed at 100%, successfully screening all 50,000 / 50,000 synthetic mixtures.
- Execution Time: The entire combinatorial matching took only 4.592 seconds on a personal computer. This ultra-fast performance is achieved by compiling the computational core into native C code prior to deployment.

3. Quantitative Composition & Spectral Decomposition

The program successfully extracted the target sample components back to its origin. The dotted lines on the plot illustrate how each individual reference standard scales to form the final multi-component mixture:

- GP1 (FePO₄) [Blue Dotted Line]: Yielded 24.73% (Fraction: 0.2473), representing the secondary phase.
- GP2 (FeS) [Orange Dotted Line]: Yielded 75.27% (Fraction: 0.7527), representing the dominant component.
- *Note: The result matches the ground truth mixture with a deviation of less than 0.3%, verifying the high numerical fidelity of the algorithm.*

4. Statistical Reliability Metrics

- R-factor: 0.00000461 (An exceptionally low value approaching 0). This numerical precision is visually confirmed on the graph, where the solid blue line (Target Sample) and the dashed red line (LCF Result) are perfectly superimposed.
- Residual [Solid Grey Line]: Located at the bottom baseline ($y = 0.0$), the residual remains completely flat, indicating a near-zero difference between the experimental data and the simulated model throughout the entire energy range.
- Chi-square: 0.002693
- Reduced Chi-square: 0.000003
- Degrees of Freedom: 800

Test Dataset & Application Deployment

To replicate this verification test or experiment with real-world case studies, users can download the official test dataset package containing both this computer-merged spectrum and physical samples prepared directly in the laboratory.

Download: **ThePrae Runtime Tests (ZIP)**

*(Note: The latest user manual and the fully compiled application are maintained on our main GitHub repository. The software is provided as a **Windows** stand-alone .exe binary that runs out-of-the-box with zero python dependencies or pip setup required. The current version has been fully verified and benchmarked on Windows 11 Home).*